

SAIKUMAR KALERU

Hyderabad, India | +91 9398320218 | saikumarkaleru56@gmail.com | [linkedin.com/in/saikumar-kaleru](https://www.linkedin.com/in/saikumar-kaleru)

PROFESSIONAL SUMMARY

Quant and market risk analyst with an MBA in Finance and a B.E. in Computer Science, NISM Series-XV certified, with close to two years on a Futures and Options trading desk at D. E. Shaw. Strong in options pricing and Greeks, volatility, and Value at Risk, and fluent in Python and SQL for building and testing quant and risk models, from implied volatility to Greek validation to VaR backtesting on real market data.

SKILLS

Derivatives & Pricing: Options and futures, Black-Scholes, binomial trees, Monte Carlo, option Greeks (delta, gamma, vega, theta), implied volatility, payoff and P&L modeling

Market & Credit Risk: Value at Risk (historical, parametric, Monte Carlo), expected shortfall, stress testing, backtesting, credit risk (Altman Z, Merton, PD, LGD, EAD), P&L attribution

Quant & Programming: Python (pandas, NumPy), SQL, vectorized backtesting, time series analysis, statistics

Markets: Nifty and Bank Nifty, open interest, volatility surface, F&O and commodity markets

Tools: Advanced Excel, Power BI, MS SQL, PostgreSQL, TradingView

EXPERIENCE

D. E. Shaw India Pvt. Ltd. | Member Technical (F&O Trading Desk)

May 2022 to Jan 2024

- Tracked implied volatility, open interest, and the full option Greeks (delta, gamma, vega, theta) across the F&O universe.
- Built analytics tools and dashboards the desk used to monitor the volatility surface, exposure, and positioning across instruments.
- Evaluated option strategies such as straddles, strangles, and spreads on a risk and return basis under different volatility views.
- Built automated Python pipelines to capture, clean, and process daily F&O market data, cutting manual work and errors.
- Prepared daily and weekly research reports on volatility and market data for the desk under tight daily deadlines.
- Validated model outputs against market data and cross checked results before they went into daily reports.

Arcesium India Pvt. Ltd. | Site Reliability Engineer (Intern)

Jan 2022 to May 2022

- Kept AWS EC2 and RDS services highly available and resolved production database issues.
- Managed and optimized MS SQL and PostgreSQL databases, improved slow running queries, and evaluated Flyway for automated schema migration.

PROJECTS

Options Pricing & Risk Analytics Engine: Black-Scholes, Binomial & Monte Carlo with Full Greeks & IV Solver | Python

- Priced options with Black-Scholes, a binomial tree, and Monte Carlo, with the full Greeks, an implied volatility solver, and a live vol smile from a real option chain.
- Checked analytic Greeks against finite difference estimates and cross checked prices across methods.
- Wrote unit tests for pricing and Greek accuracy and packaged the model as a reusable Python module.

Multi Leg Options Strategy Analyzer: Payoff, P&L, Net Greeks & Probability of Profit | Python

- Modeled multi-leg strategies with payoff, P&L, net Greeks, and probability of profit over a spot and vol grid.
- Compared strategies on a like for like risk basis to guide selection under different volatility views.
- Charted payoff and P&L across a spot and volatility grid and packaged the strategies as a reusable Python module.

Portfolio Market Risk Engine: Historical, Parametric & Monte Carlo VaR with Expected Shortfall & Backtesting | Python

- Computed one day VaR and expected shortfall using historical, parametric, and Monte Carlo methods.
- Backtested VaR against realized moves to check the model held up.
- Added component and marginal VaR and correlated Monte Carlo with a Cholesky factorization.
- Added unit tests and a clean report comparing the three VaR methods side by side.

EDUCATION

MBA in Finance, University College of Commerce & Business Management, Hyderabad

2024 to 2026

CGPA 8.9

B.E. Computer Science & Engineering, Chaitanya Bharathi Institute of Technology, Hyderabad

2018 to 2022

CGPA 8.8

RELEVANT COURSEWORK

Financial Derivatives, Risk Management, Quantitative Techniques, Security Analysis & Portfolio Management, Financial Modeling & Valuation, Business Statistics

CERTIFICATIONS

NISM Series-XV: Research Analyst (SEBI/NISM), Reg. No. NISM-202600123995

ACHIEVEMENTS

- UGC NET 2025 (Management): 95.6 percentile.